SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _ J Jeshwanth(1921260264)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Artificial Intelligence – Search Algorithms and Intelligent Agents

1. Hospital Doctor–Shift Assignment Using Backtracking Search

Given:

-

Doctors: D1, D2, D3

Shifts: Morning (M), Afternoon (A), Night (N)

Constraints

D1 cannot work Night.

D2 must work before D3.

Only one doctor per shift.

D3 cannot work Morning.

Every doctor must be assigned exactly one shift.

Step-by-step backtracking

We assign doctors to shifts one at a time and check the constraints at every step.

Step	Assignment	Constraint check	Decision
1	D1 → Morning	D1 cannot work Night; satisfied	Accept
2	D2 → Afternoon	D2 is before D3 only if D3 is Night	Accept
3	D3 → Night	D3 cannot work Morning; satisfied	Accept
4	Check all shifts	One doctor per shift; all doctors assigned	Success

Final valid schedule

Shift	Assigned doctor
Morning	D1
Afternoon	D2
Night	D3

Conclusion: The assignment satisfies all five constraints, so backtracking terminates successfully without needing to undo any assignment.

2. Robot Navigation in a 5×5 Grid Using Search Algorithms

Objective: Find the shortest path from Start (S) to Goal (G), avoiding obstacles.

Given constraints

Grid size: \(5 \times 5\)

Allowed movements: Up, Down, Left, Right

Cost per movement: 1

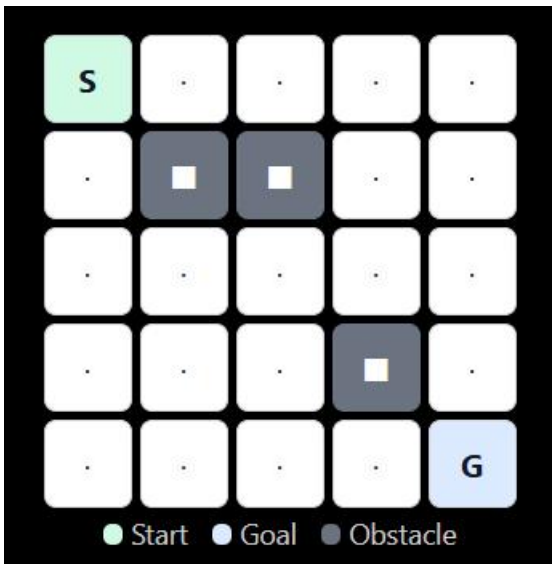
Obstacles cannot be crossed.

Manhattan distance is the heuristic.

Important: The question does not specify the exact locations of S, G, and the obstacles. The following is an illustrative example with assumed positions.

Example grid

Assume S = (1,1), G = (5,5), and obstacles at (2,2), (2,3), and (4,4).



Start
Goal
Obstacle

Manhattan distance heuristic

The heuristic estimates the distance to the goal without considering obstacles.

\[h(n)=|x_n-x_G|+|y_n-y_G| \]

For example, from S = (1,1) to G = (5,5):

$$ h(S)=|1-5|+|1-5|=8 $$

BFS versus A*

> BFS (Breadth-First Search): Uses a FIFO queue and explores nodes level by level. Since all moves cost 1, it finds a shortest path.
>
> A Search:* Uses a priority queue ordered by $f(n)=g(n)+h(n)$, where $g(n)$ is the cost so far and $h(n)$ is Manhattan distance.

Since the question asks for both a heuristic and priority-queue updates, the following uses A* to demonstrate those requirements.

Step-by-step A* search

Assume ties are broken in favor of the node with the smaller heuristic, and neighbors are considered Right, Down, Left, Up.

Step	**Node selected**	**$g(n)$**	**$h(n)$**	**$f(n)=g+h$**
1	S (1,1)	0	8	8
2	(1,2)	1	7	8
3	(1,3)	2	6	8
4	(1,4)	3	5	8
5	(1,5)	4	4	8
6	(2,5)	5	3	8
7	(3,5)	6	2	8
8	(4,5)	7	1	8
9	G (5,5)	8	0	8

Priority queue updates: At each step, the newly discovered safe neighbors are inserted into the priority queue according to their f-values. The node with the smallest f-value is expanded next. Nodes that are obstacles are never inserted.

Optimal path

$$ \begin{aligned} S &\rightarrow (1,2)\rightarrow(1,3)\\ &\rightarrow(1,4)\rightarrow(1,5)\\ &\rightarrow(2,5)\rightarrow(3,5)\\ &\rightarrow(4,5)\rightarrow G \end{aligned} $$

3. Autonomous Rescue Robot in a Partially Observable Environment

a) Analysis of the constraints

Constraint	Effect on decision-making
Four-directional movement	The robot can move only up, down, left, or right.
Obstacles	Blocked cells must be avoided.
Limited sensing	The robot can observe adjacent cells but cannot know the entire grid initially.
Movement cost = 1	Each ordinary movement adds one unit to the path cost.
Risky-zone cost = 2 extra	Entering a risky cell costs \(1+2=3\).
Minimize total cost	The robot must balance distance, safety, and risk.
Dynamic knowledge	The robot must update its internal map whenever new information is obtained.

b) Proposed solution: Online Search Agent with Uniform Cost Search

Uniform Cost Search (UCS) is suitable for finding a minimum-cost path when movement costs vary. Because the robot cannot observe the entire environment in advance, it should combine UCS with an online search strategy.

The total cost of a path is:

\[g(n)=\text{ordinary movement costs}+\text{risk penalties} \]

Steps involved

Initialize: Mark the starting cell S as known and set its cost to zero.

Sense: Observe the four adjacent cells to identify blocked, safe, or risky cells.

Update the map: Add newly discovered information to the robot's internal representation.

Expand the lowest-cost node: Use UCS to select the reachable known cell with the lowest accumulated cost.

Avoid obstacles: Never traverse blocked cells.

Account for risk: Add 3 for entering a risky cell and 1 for an ordinary safe cell.

Move and observe: Travel toward the selected cell and sense the environment again.

Replan: If a new obstacle or risky zone is discovered, update the map and calculate a new route.

Terminate: Stop when the survivor's location G is reached.

UCS pseudocode

```
START
frontier = priority_queue()
frontier.push(S, cost=0)
cost[S] = 0
WHILE frontier is not empty:
    current = frontier.pop_lowest_cost()
    IF current == G:
        RETURN reconstruct_path(current)
    OBSERVE adjacent cells
    UPDATE internal map
    FOR each known neighbor of current:
        IF neighbor is not blocked:
            move_cost = 1
            IF neighbor is risky:
                move_cost = 3
            new_cost = cost[current] + move_cost
            IF neighbor not visited OR new_cost < cost[neighbor]:
                cost[neighbor] = new_cost
                frontier.push_or_update(neighbor, new_cost)
    IF no safe known move is available:
        EXPLORE another reachable known cell
RETURN "No path currently known"
```

The robot should replan whenever newly observed information changes the best route. In a genuinely online setting, it may need to explore unknown cells before it can establish that no route exists.

c) Application of AI concepts

1. Intelligent agent

The robot senses the environment, processes information, plans routes, and performs actions to reach the survivor. It uses sensors for perception and a control system for movement.

2. Rationality

A rational robot chooses actions that maximize expected mission success while minimizing travel cost and exposure to danger, based on its available information.

For example, it may choose a slightly longer safe route instead of a shorter route containing dangerous cells.

3. Environment type

Property	Classification	Reason
Observability	Partially observable	Only adjacent cells can be sensed.
Determinism	Potentially stochastic	Unknown conditions may make outcomes uncertain.
Task structure	Sequential	Each movement affects later decisions.
Dynamics	Dynamic	Newly discovered obstacles or hazards change the map.
State representation	Discrete	The building consists of grid cells.
Agents	Single-agent	The robot performs the navigation task independently.

d) Why is this approach suitable?

UCS finds a minimum-cost path when all relevant traversal costs are known.

Online search enables decisions when the complete environment is initially unknown.

Dynamic replanning allows the robot to respond to newly discovered obstacles.

Cost-aware decisions account for both ordinary movement and risky zones.

Safety-aware planning helps the robot avoid known hazards while reaching the survivor.

Conclusion: An online search agent using UCS is appropriate because it combines exploration and replanning with minimum-cost path selection. Manhattan distance can also be incorporated into an A* variant, but UCS alone does not use a heuristic.